

# LAB MID



**Submitted by:**

| Name              | Registration No. |
|-------------------|------------------|
| Muhammad Abdullah | FA24-BSE-042     |

**Submitted to: Anayat Ullah**

**Course: Information security**

**Dated: April 6, 2026.**

---

**DEPARTMENT OF COMPUTER SCIENCE  
COMSATS UNIVERSITY ISLAMABAD,  
ISLAMABAD CAMPUS**

## QUESTION 1:

Hashtype:

32 Characters = MD5

```
C:\Users\      \Desktop\hashcat\hashcat-7.1.2>hashcat -a 0 -m 1400 5f4dcc3b5aa765d61d8327deb882cf99 rockyou-30000.rule
hashcat (v7.1.2) starting

OpenCL API (OpenCL 3.0 ) - Platform #1 [Intel(R) Corporation]
=====
* Device #01: Intel(R) UHD Graphics, 1635/3270 MB (817 MB allocatable), 8MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

Hash '5f4dcc3b5aa765d61d8327deb882cf99': Token length exception

* Token length exception: 1/1 hashes
  This error happens if the wrong hash type is specified, if the hashes are
  malformed, or if input is otherwise not as expected (for example, if the
  --username or --dynamic-x option is used but no username or dynamic-tag is present)

No hashes loaded.

Started: Mon Apr 06 15:06:06 2026
Stopped: Mon Apr 06 15:06:11 2026
```

Dictionary Attack:

```
5f4dcc3b5aa765d61d8327deb882cf99.password

Session.....: hashcat
Status.....: Cracked
Hash.....: 0 (MD5)
Hash.Target.....: 5f4dcc3b5aa765d61d8327deb882cf99
Time.Started.....: Mon Apr 06 15:11:46 2026 (0 secs)
Time.Estimated...: Mon Apr 06 15:11:46 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.....: File (rockyou.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#01.....: 501 H/s (0.03ms) @ Accel:1024 Loops:1 Thr:63 Vec:1
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 5/5 (100.00%)
Rejected.....: 0/5 (0.00%)
Restore.Point....: 0/5 (0.00%)
Restore.Sub.#01..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#01...: Password -> comsats
```

Brute Force Attack:

```
C:\Users\      \Desktop\hashcat\hashcat-7.1.2>hashcat -m 0 -a 3 hash.txt ?l?l?l?l?l?l --show
2bf44cfc1766c957a59f85e1d7fa5e11:Aimal
```

## Combination Attack:

```
Recovered.....: 0/1 (0.00%) Digests (total), 0/1 (0.00%) Dig  
Progress.....: 9/9 (100.00%)  
Rejected.....: 0/9 (0.00%)  
Restore.Point....: 3/3 (100.00%)  
Restore.Sub.#01...: Salt:0 Amplifier:2-3 Iteration:0-2  
Candidate.Engine.: Device Generator  
Candidates.#01...: hellosats -> comsats
```

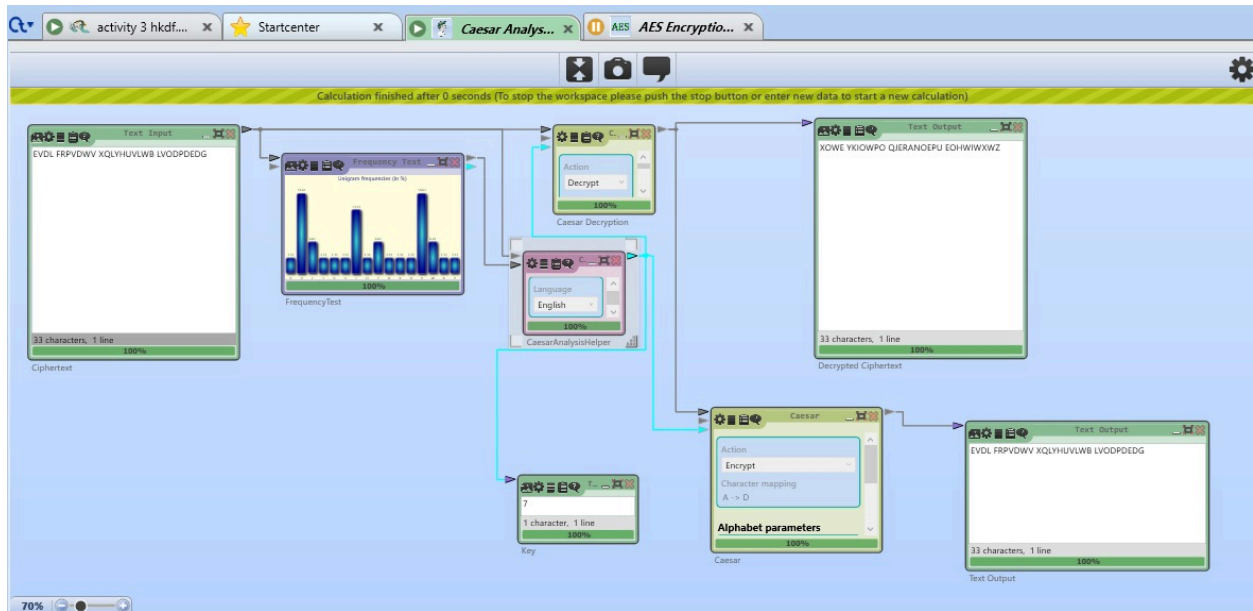
## Cracked passwords:

```
Candidate.Engine.: Device Generator  
Candidates.#01...: Password -> comsats
```

## QUESTION 2:

### 2.1:

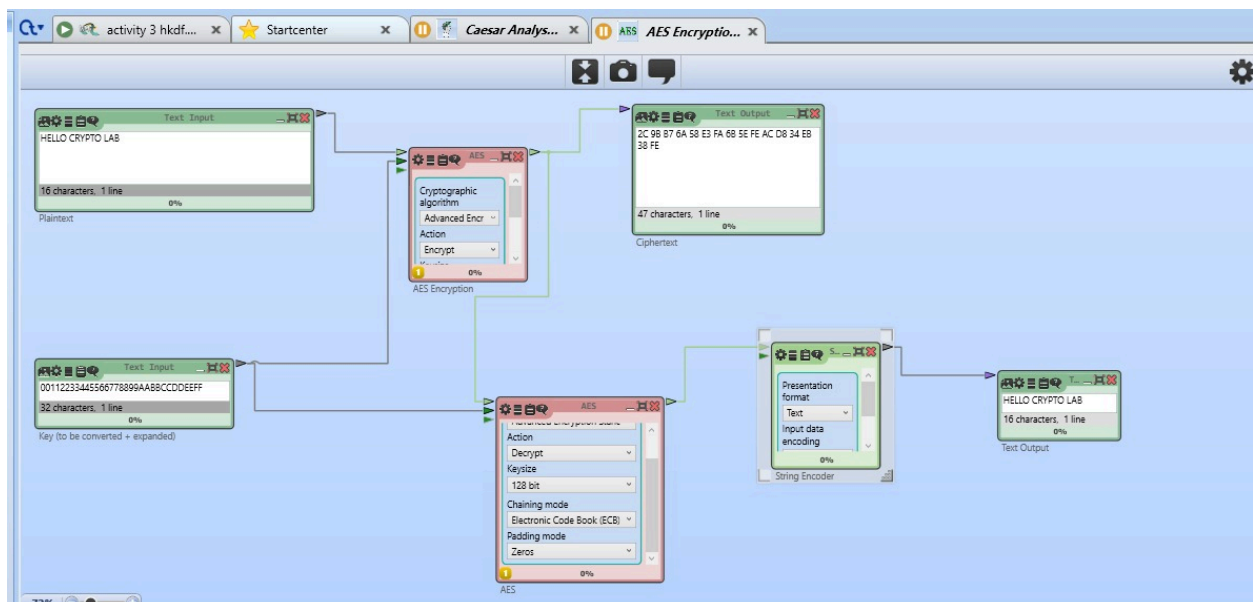
#### Working & VERIFICATION:



most likely key values are v and d

### 2.2:

#### Encryption & Decryption:



## QUESTION 3:

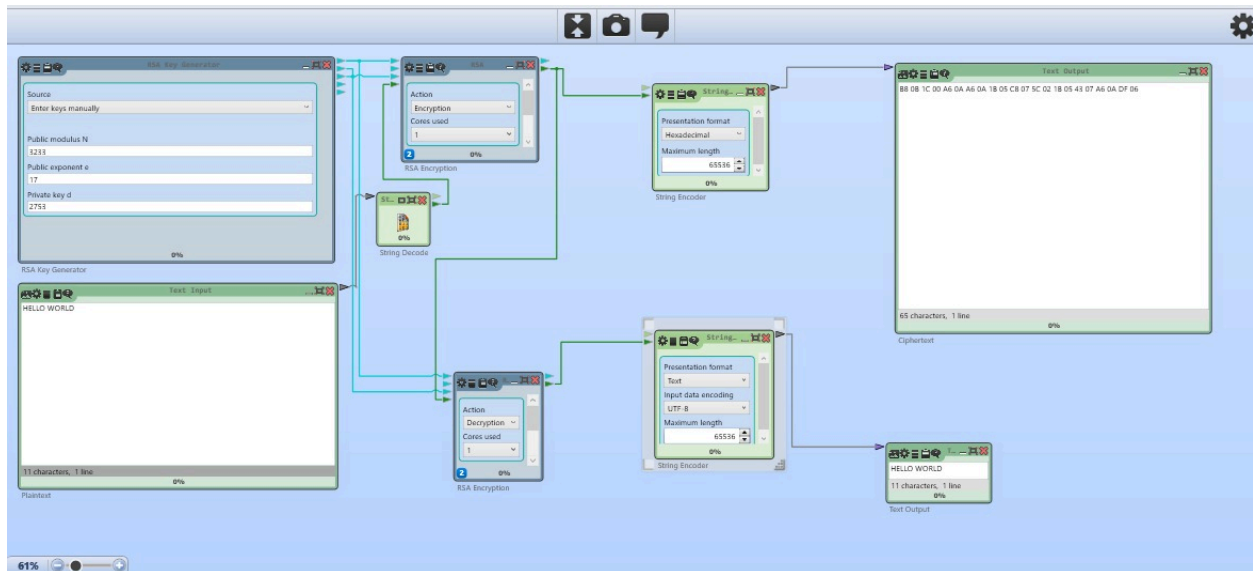
```
muhammadabbas@DESKTOP-D1ND00U:~$ ./fastcoll -o file1.bin file2.bin
MD5 collision generator v1.5
by Marc Stevens (http://www.win.tue.nl/hashclash/)

Using output filenames: 'file1.bin' and 'file2.bin'
Using initial value: 0123456789abcdeffedcba9876543210

Generating first block: .....
Generating second block: 500.....
Running time: 1.84216 s
muhammadabbas@DESKTOP-D1ND00U:~$ md5sum file1.bin file2.bin
6b40b2d6b24705cf2f89211814981d27 file1.bin
6b40b2d6b24705cf2f89211814981d27 file2.bin
muhammadabbas@DESKTOP-D1ND00U:~$ hexdump -C file1.bin | head
xdump -C file2.bin | head00000000 94 01 5a ea 6a 36 0d 71 2b af 01 c7 0a dc 5f 96 |..Z.j6.q+...._|
00000010 f0 b8 a8 63 ea d6 30 ef 26 4a 82 5a f4 53 ad 1a |...c...0.&J.Z.S..|
00000020 ab ed ec e4 60 03 26 3b eb ff f2 3a f4 65 01 65 |....`.&;...:e.e|
00000030 27 8f 44 22 62 41 16 d5 5b dc 96 47 a5 32 96 8e |'.D"bA..[.G.2..|
00000040 7b 99 cb bd 5b 54 f8 d9 f1 66 c8 0e 3f d6 71 ee |{...[T...f...q.|
00000050 48 88 8d 03 0b 79 61 1a cd 57 f3 cc 59 31 59 22 |H....ya..W..Y1Y"|
00000060 7d 27 57 e2 29 89 83 f1 75 4d d4 cf 9c be 7e e8 |}'W.)...uM....~|
00000070 d1 eb 0e 12 00 b3 b4 e0 cd 5e c8 40 f9 c8 94 f2 |.....^.@....|
00000080
00000080
muhammadabbas@DESKTOP-D1ND00U:~$ hexdump -C file2.bin | head^C
muhammadabbas@DESKTOP-D1ND00U:~$ ^C
muhammadabbas@DESKTOP-D1ND00U:~$ ^C^C
muhammadabbas@DESKTOP-D1ND00U:~$ ^C
muhammadabbas@DESKTOP-D1ND00U:~$ ^C
muhammadabbas@DESKTOP-D1ND00U:~$ nano file1.bin
muhammadabbas@DESKTOP-D1ND00U:~$ nano file2.bin
muhammadabbas@DESKTOP-D1ND00U:~$ md5sum file1.bin file2.bin
b39b547c3282471c8574be78d89c97e2 file1.bin
60d859586eaa59151ee0d22026e4d1d3 file2.bin
muhammadabbas@DESKTOP-D1ND00U:~$ hexdump -C file1.bin | head
00000000 64 73 61 64 94 01 5a ea 6a 36 0d 71 2b af 01 c7 |dsad..Z.j6.q+...|
00000010 0d dc 5f 96 f0 b8 a8 63 ea d6 30 ef 26 4a 82 5a |...c...0.&J.Z|
00000020 f4 53 ad 1a ab ed ec e4 60 03 26 3b eb ff f2 3a |.S.....`.&;...:|
00000030 f4 65 01 65 27 8f 44 22 62 41 16 d5 5b dc 96 47 |.e.e'.D"bA..[.G|
00000040 a5 32 96 8e 7b 99 cb bd 5b 54 f8 d9 f1 66 c8 0e |.2..{...[T...f..|
00000050 3f d6 71 ee 48 88 8d 03 0b 79 61 1a cd 57 f3 cc |?.q.H....ya..W..|
00000060 59 31 59 22 7d 27 57 e2 29 89 83 f1 75 4d d4 cf |Y1Y"}'W.)...uM..|
00000070 9c be 7e e8 d1 eb 0e 12 00 b3 b4 e0 cd 5e c8 40 |..~.....^.@|
00000080 f9 c8 94 f2 0d |....|
00000085
```

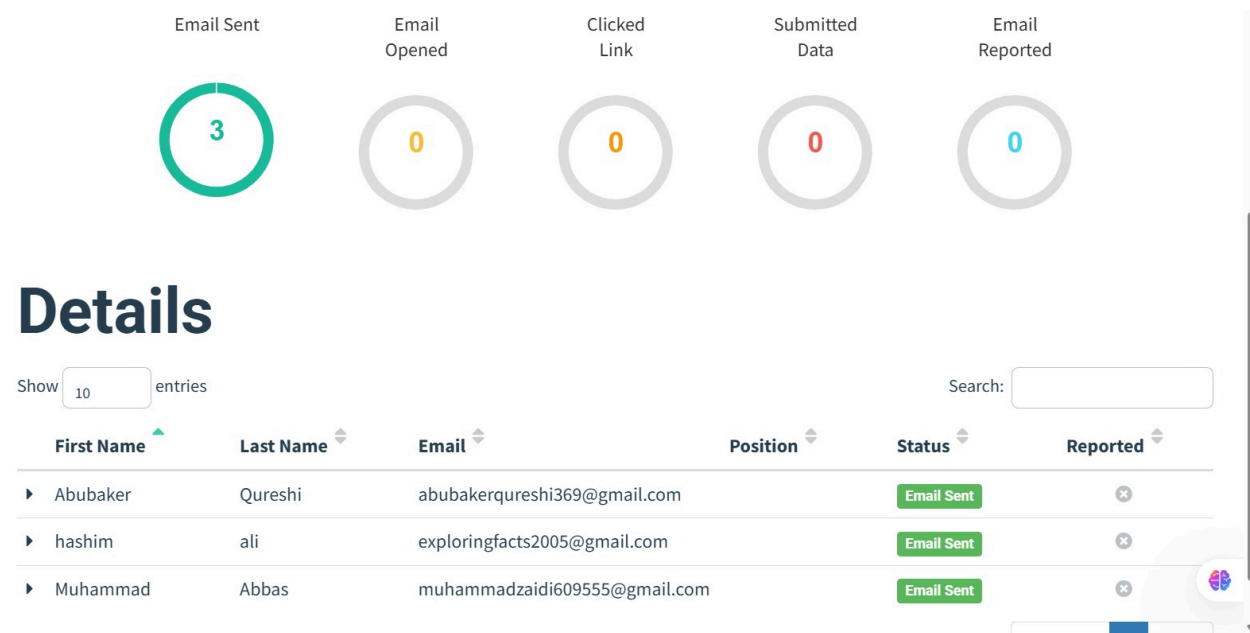
## QUESTION 4:

### Encryption & Decryption:



# QUESTION 5:

Campaign results:



Details

Show 10 entries

Search:

| First Name | Last Name | Email                         | Position | Status     | Reported |
|------------|-----------|-------------------------------|----------|------------|----------|
| ▶ Abubaker | Qureshi   | abubakerqureshi369@gmail.com  |          | Email Sent | ×        |
| ▶ hashim   | ali       | exploringfacts2005@gmail.com  |          | Email Sent | ×        |
| ▶ Muhammad | Abbas     | muhammadzaidi609555@gmail.com |          | Email Sent | ×        |

